

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$M_{0c}^a M_{0b}^c = (M_0^2)_b^a = \begin{pmatrix} -\frac{\alpha'}{2}[A_\mu, A_\nu]\psi^\mu\psi^\nu + \bar{T}T & -i\sqrt{\alpha'}(\bar{T}A'_\mu - A_\mu\bar{T})\psi^\mu \\ -i\sqrt{\alpha'}(TA'_\mu - A'_\mu T)\psi^\mu & -\frac{\alpha'}{2}[A'_\mu, A'_\nu]\psi^\mu\psi^\nu + T\bar{T} \end{pmatrix}.$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \|r_h^+ u - r_h u\|_{0,2}^2 &= a_h^+(r_h^+ u - r_h u, w) \\ &= a_h^+(r_h^+ u - r_h u, w - w_h^+) + a_h^+(r_h^+ u - r_h u, w_h^+) \\ &= a_h^+(r_h^+ u - r_h u, w - w_h^+) + a_h^+(u - r_h u, w_h^+) \\ &= a_h^+(r_h^+ u - r_h u, w - w_h^+) + a_h^+(u - r_h u, w_h^+ - w_h) \\ &\quad + a_h^+(u - r_h u, w_h) - a_h(u - r_h u, w_h) \\ &=: T_1 + T_2 + T_3, \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \sum_{n=1}^{\infty} c_{1,n} T_1^n (\{x_\tau^0\}_{\tau=1}^r) &\equiv \sum_{n=1}^{\infty} c_{1,n} T_1^n (\{x_\tau^0 \tilde{f}_0^n\}_{\tau=1}^r), \\ \sum_{n=1}^{\infty} c_{2,n} T_2^n (\{x_\theta^0\}_{\theta=r+1}^{r+s}) &\equiv \sum_{n=1}^{\infty} c_{2,n} T_2^n (\{x_\theta^0 \tilde{f}_0^n\}_{\theta=r+1}^{r+s}). \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} &\|\psi_{n,t}(b)^* \psi_{n,t}(b) + (\psi_{n,e}(c) + \alpha 1_{k_n})^* (\psi_{n,e}(c) + \alpha 1_{k_n}) - \|b\|^2 1_{k_n}\| \\ &\leq \|\psi_{n,t}(b)^* \psi_{n,t}(b) - \psi_{n,e}(b^* b)\| + \|\psi_{n,e}(c)^* \psi_{n,e}(c) - \psi_{n,e}(c^* c)\| \\ &= \|\psi_{n,t^{-1}}(b^*) \psi_{n,t}(b) - \psi_{n,e}(b^* b)\| + \|\psi_{n,e}(c^*) \psi_{n,e}(c) - \psi_{n,e}(c^* c)\| < 2\varepsilon_n \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

- (i) $B = B_1 \cup B_2, \tilde{B} = E_1 \cup E_2$ and $\omega(E_i) \geq \frac{1}{2}\omega(\tilde{B}), i = 1, 2$;
- (ii) $b(x) - b(y)$ does not change sign for all $(x, y) \in B_i \times E_i, i = 1, 2$;
- (iii) $|b(x) - m_b(\tilde{B})| \leq |b(x) - b(y)|$ for all $(x, y) \in B_i \times E_i, i = 1, 2$.